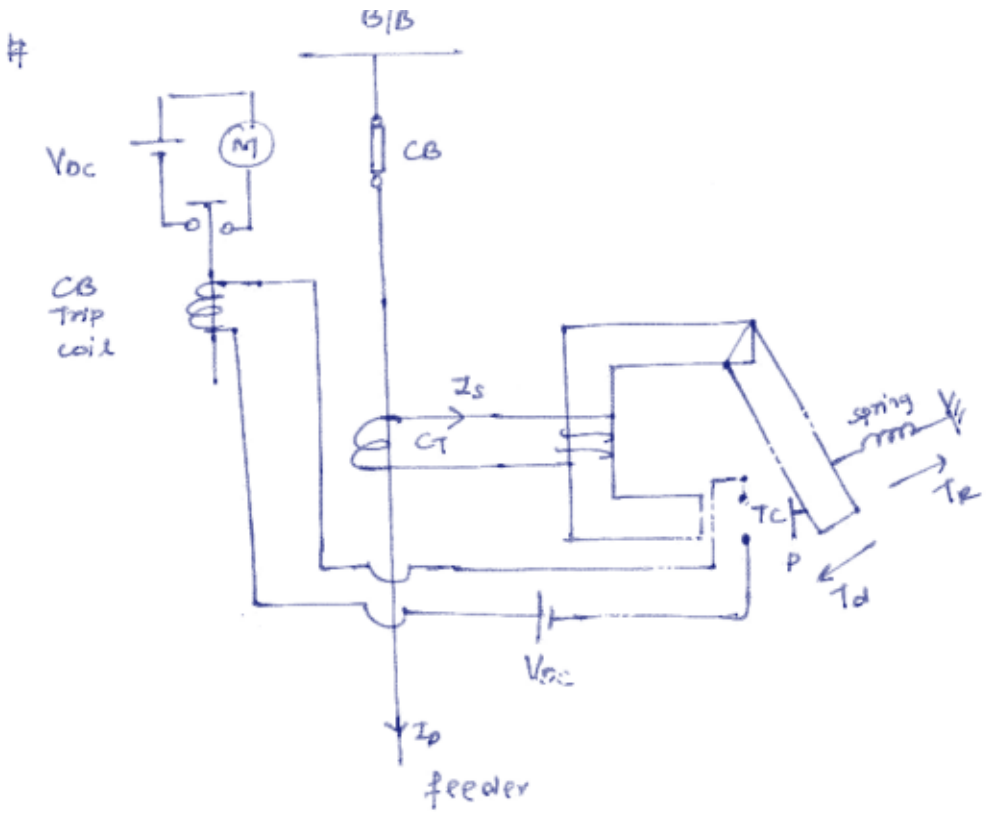
Circuit Breakers

Following sequence is used after attending maintenance, before charging of equipment :

(1) Open Earth Switch

(2) Close Isolator

(3) Close CB.



Bus (B/B) feeder with CB; CT secondary current Is operating an electromagnetic relay (plunger P, spring, trip contacts TC, torques Td and TR); DC supply Vdc feeds CB trip coil and motor (M) for the breaker; feeder current Ip

T_R -> Restraining or control Torque ; T_d -> Deflecting Torque ; TC -> Trip contacts ; P -> plunger

(1) No fault (Block) - $T_d \leq T_R$: TC - open , CB - close

(2) Fault ($T_d > T_R$) : TC - close , CB - open

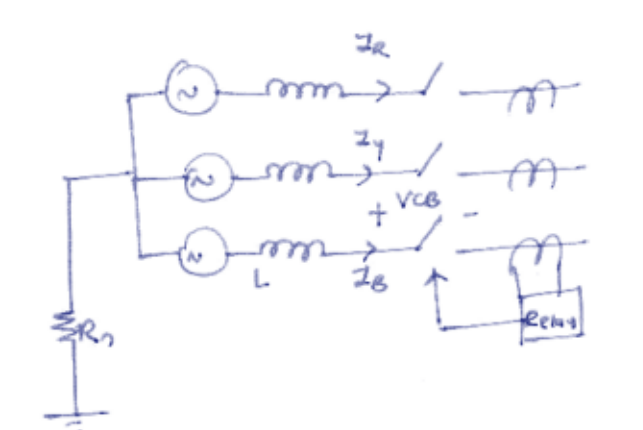
CB Circuit Breaker :

* Circuit Breaker is used to isolate faulty sections from the healthy sections. It is filled with arc quenching medium like SF6, vacuum.
* SF6 breakers are used for 33 kV vtg and above vtg level.
* Vacuum breakers are used for below 33 kV vtg level.
* Reasons for arc :
  (1) Thermal emissions : If contact surfaces of breaker are separated, surface area is reduced causing high current density, so that very high heat is developed which initiates arc.
  (2) Secondary emissions : Fast moving carriers collide with surrounding molecules and liberate more carriers which continues the arc.
  (3) Field emissions : Due to energy exchange between L and C components, huge electric field is developed across the breaker poles which will re-initiate the arc, which is called re-striking arc due to field emissions.

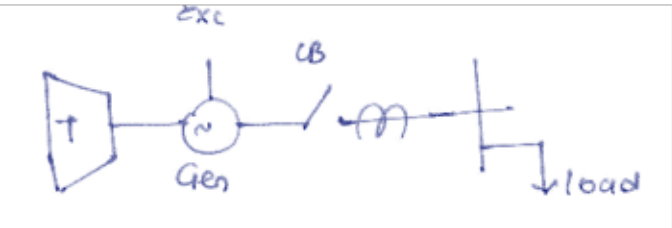
Methods of arc quenching :

(1) Natural current zero interruption : It is used in all HVAC breakers. For 50 Hz frequency signal, current is passing through the natural zero point 2 times in each cycle. At this point magnitude of the current is less and can be easily quenched.
(2) Force current zero interruption OR Current chopping : It is used in low magnetizing AC current interruptions.
(3) High Resistance Interruption : It is used in DC system where natural current zero is not available.

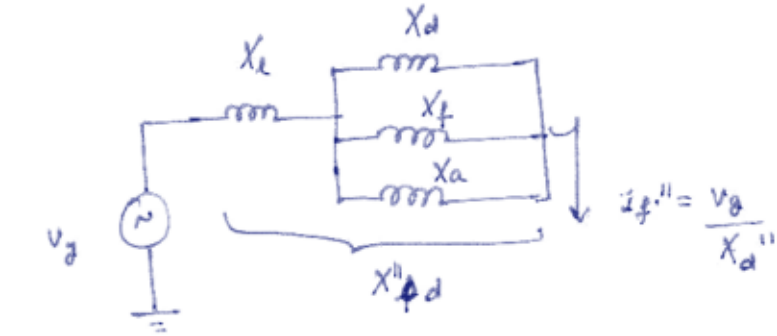
Natural current zero Interruption :



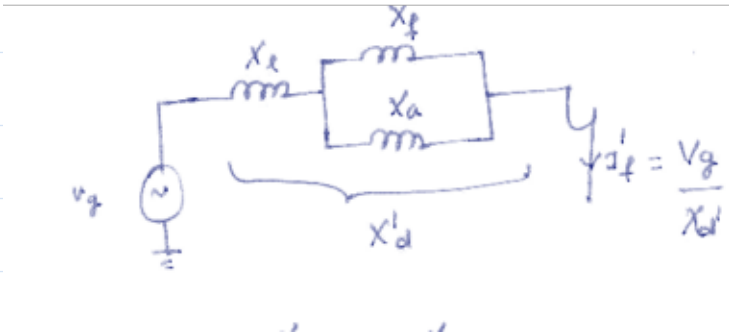
Three-phase generator sources with series inductance L, phase currents IR, IY, IB, circuit breaker with voltage VCB across it, relay, neutral earthed through Rn



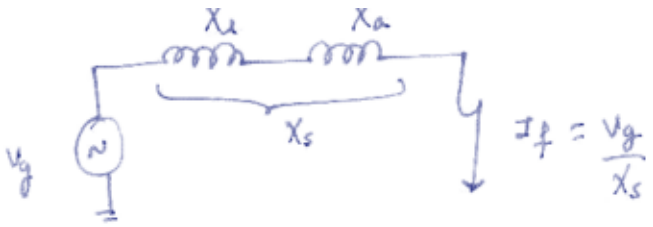
Turbine, generator with excitation (Exc), CB, line inductance to load bus



Sub-transient equivalent: Vg behind leakage reactance Xl feeding parallel Xd (damper), Xf (field), Xa (armature); X"d total; If" = Vg / X"d



Transient equivalent: Xl in series with Xf // Xa giving X'd; If' = Vg / X'd



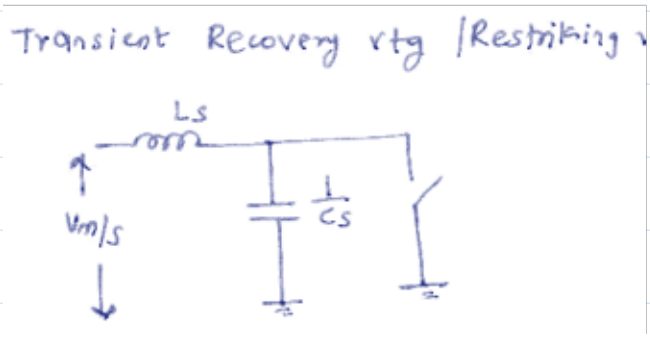
X_l, X_d, X_f, X_a – leakage, Damper, field armature reactances.

Steady state: Xl in series with Xa giving Xs; If = Vg / Xs; Xl, Xd, Xf, Xa - leakage, Damper, field, armature reactances

$X''_d, X'_d, X_s, I''_f, I'_f, I_f$ = Subtransient, transient, steady state reactances and fault currents

$$X''_d < X'_d < X_s; \quad I''_f > I'_f > I_f$$

(1) Transient Recovery vtg / Restriking vtg (TRV)



Source Ls with shunt capacitor Cs across the breaker (earthed): circuit for TRV

$$V_c = V_{CB} = \frac{V_m/s}{\left(Ls + \frac{1}{Cs}\right)} = \frac{V_m}{s(LCs^2 + 1)}$$

$$TRV = V_c(t) = V_m[1 - \cos(\omega_n t)]$$

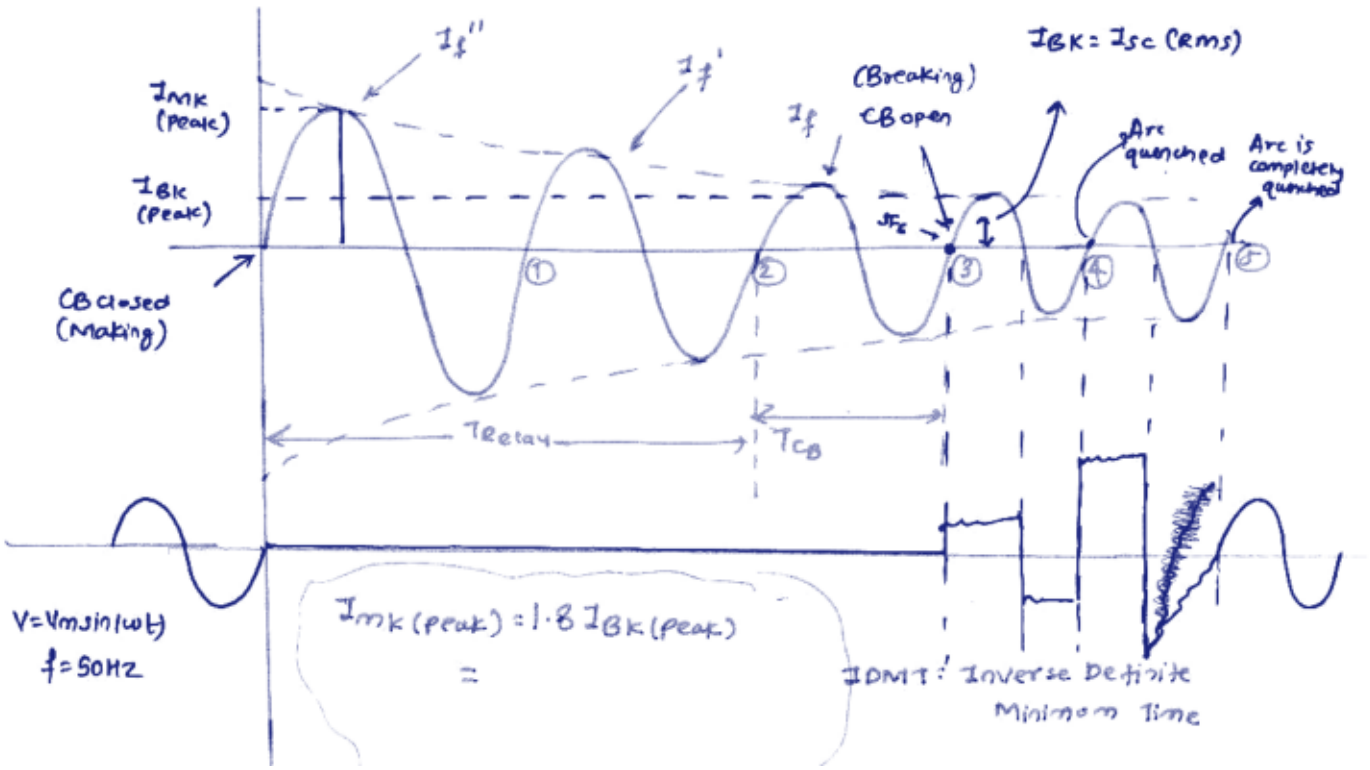
$\omega_n = 2\pi f_n = \frac{1}{\sqrt{LC}}$ = Natural freq of oscillation

$$(TRV)_{max} = 2V_m = 2\sqrt{2}\, V_{ph} = 2\sqrt{2}\,\frac{V_L}{\sqrt{3}}$$

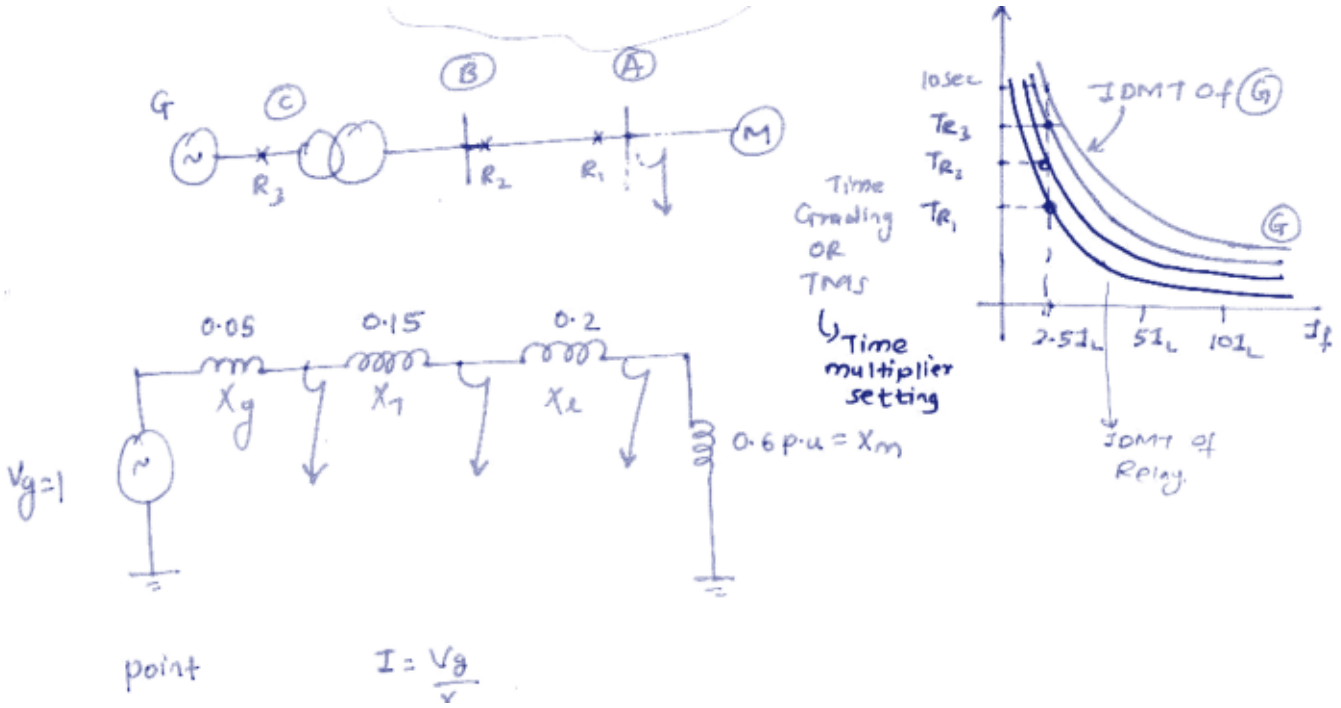
$$(TRV)_{max} = 1.632\, V_{L(rms)}$$

occurs at $\omega_n t = \pi$

$$t = \frac{\pi}{\omega_n}$$



Fault current waveform with making (CB closed, peak IMK), breaking (peak IBK, IBK = Isc rms), If'', If', If envelope; relay time and CB time; arc quenched at current zero; below it the voltage V = Vm sin(wt), f = 50 Hz and the arc voltage; IMK(peak) = 1.8 IBK(peak)



Radial system G - C - B - A - M with relays R3, R2, R1 and fault points A, B, C; reactances Xg = 0.05, X1 = 0.15, X2 = 0.2, motor 0.6 p.u, Vg = 1; time-current IDMT curves for relays and generator (10 sec; TR1, TR2, TR3; 2.5 IL, 5 IL, 10 IL) showing time grading / time multiplier setting (TMS)

$I_{MK(peak)} = 1.8 I_{BK(peak)}$; IDMT : Inverse Definite Minimum Time

Point / $I = \frac{V_g}{X}$

No fault : 1 p.u = I_L = Full load current

(A) : 2.5 p.u = 2.5 I_L

(B) : 5 p.u = 5 I_L

(C) : 20 p.u = 20 I_L

(1) Nominal current : Rated current flowing through the Breaker poles at rated vtg is called Nominal current.

(2) Nominal vtg : Rated vtg at which CB is designed to operate.

(3) Short Circuit OR Breaking current : The RMS value of the current passing through the breaker poles during opening of the Bkr. This corresponds to fault current of the system. Normal current rating of this parameter is 25, 40, 50, 63 kA are practically used.

(4) Short Circuit Capacity : The MVA capacity of the breaker corresponding to the short circuit current at rated vtg is short circuit capacity of the Breaker.

$$(MVA)_{S.C} = \sqrt{3} \cdot V_L \cdot I_{BK}$$

(5) Making current : Peak value of the current passing through Bkr poles at initial closing time of the Bkr is making current ($I_{MK\,peak}$).

$$I_{MK(peak)} = 2.55\, I_{BK(rms)}$$

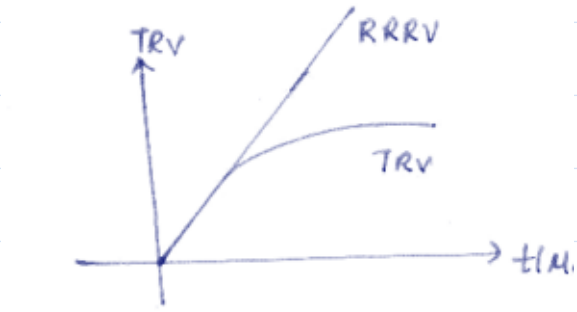
(6) Breakers are designed to withstand this $I_{MK\,(peak)}$ value.

(7) S.C Making current : MVA rating, if peak value of the current passing through the Bkr poles at rated vtg corresponds to the making current.

$$(MVA)_{peak} = 2.55\,(MVA)_{S.C}$$

Rate of Rise of Restriking vtg (RRRV)

The rate at which restriking vtg is rising is expressed by RRRV. These correspond to switching O.V. Insulators are designed to withstand these vtg.



TRV and its tangent RRRV against t (microseconds): TRV rises and levels off, RRRV is the slope

$$RRRV = \frac{d}{dt}(TRV) = \frac{d}{dt}[V_m(1-\cos\omega_n t)]$$

$$RRRV = V_m \omega_n \sin(\omega_n t) \quad kV/\backslash mu\ sec$$

$$(RRRV)_{max} = V_m \omega_n$$

occurs at $\omega_n t = \pi/2$, $t = \frac{\pi}{2\omega_n}$

$$(\text{Avg RRRV}) = \frac{(TRV)_{max}}{\text{Time at which } (TRV)_{max} \text{ occurs}} = \frac{(TRV)_{max}}{(\pi/\omega_n)}$$

Operating sequence or Duty cycle :

O - 0.3 sec - CO - 3 min - CO ; O - open ; CO - Close followed by open.

* In case of auto reclosing, CB operating sequence is practically used for improving the system stability.

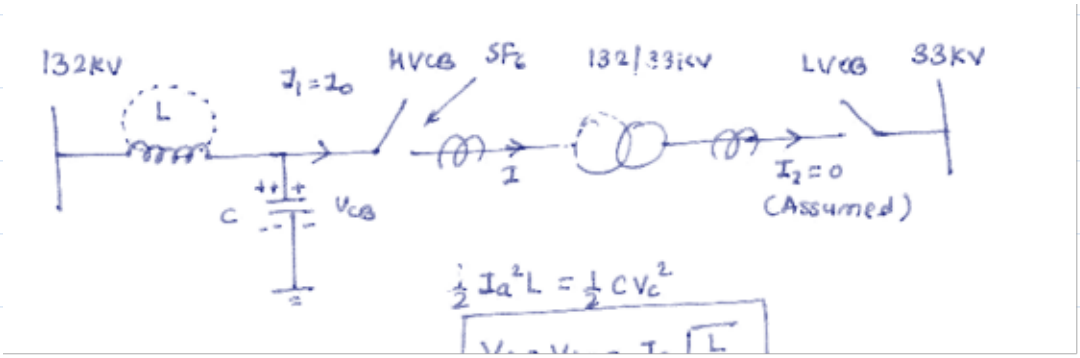
* If fault occurs in one of the phases then corresponding CB is initially open and waiting for minimum time of 0.3 sec (Dead time), and now breaker is closed after this dead time assuming that temporary fault in the system.

If fault still persists, immediately open the CB and wait up to 3 minutes (Re-claim time), then close the breaker to improve stability. If still fault exists then open the breaker immediately and declare as the permanent fault.

DAY # 11 [19/9/25]

Forced current zero interruption or current chopping :

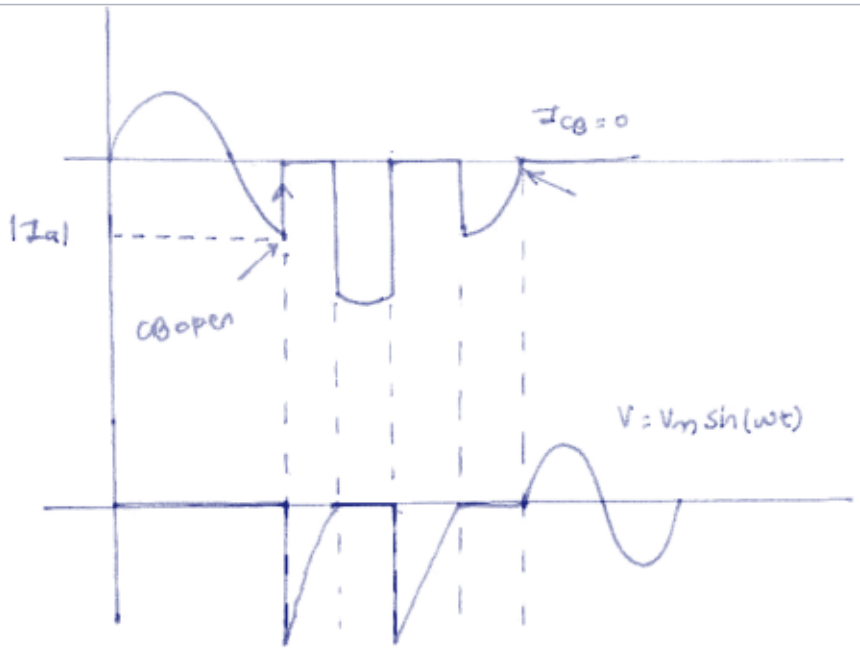
-> It is used to interrupt the current in the low magnetizing current applications like no load current of power X'mer.



132 kV bus with line inductance L and capacitance C (voltage VCB), HV CB (SF6) feeding 132/33 kV transformer, LV CB open (I2 = 0, assumed), 33 kV bus; chopping current Ia

$$\frac{1}{2} I_a^2 L = \frac{1}{2} C V_c^2$$

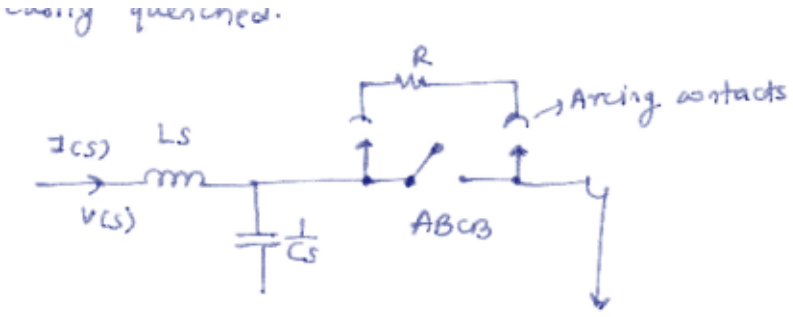
$$V_c = V_{CB} = I_a \sqrt{\frac{L}{C}}$$



Current waveform chopped at |Ia| when CB opens (ICB = 0 afterwards) and the resulting voltage V = Vm sin(wt) with oscillation after chopping

Resistance switching :

In case of Air Blast CB, resistance switching is used in which an additional resistance is connected in parallel to the CB, which will divert the path of current so that magnitude of current passing through the breaker is reduced, and hence arc can be easily quenched.



Source Ls with Cs; air blast circuit breaker ABCB with arcing contacts and parallel resistance R

Resistance switching :

$$Z(s) = Ls + \frac{R \times \frac{1}{Cs}}{R + \frac{1}{Cs}} = Ls + \frac{R}{RCs+1} = \frac{RLCs^2 + Ls + R}{RCs+1}$$

$$I(s) = \frac{V(s)}{Z(s)} = \frac{V(s)(1+RCs)}{RLCs^2 + Ls + R}$$

Ch. eqn : $1 + GH = 0 : RLCs^2 + Ls + R ; RLC\left[s^2 + \frac{1}{RC}s + \frac{1}{LC}\right] = 0$

For second order : $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-\frac{1}{RC} \pm j\sqrt{\frac{4}{LC} - \left(\frac{1}{RC}\right)^2}}{2} = -\sigma \pm j\omega_d$$

Damped freq of oscillations :

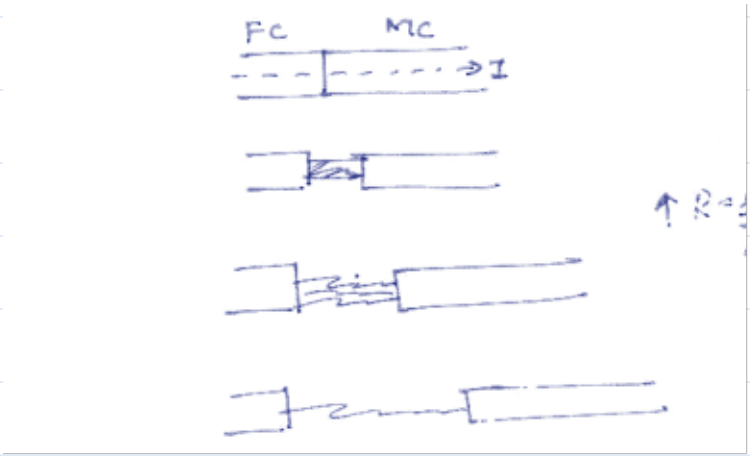
$$\omega_d = \frac{1}{2}\sqrt{\frac{4}{LC} - \left(\frac{1}{RC}\right)^2} \quad \text{rad/sec}$$

For $\omega_d = 0 : \frac{4}{LC} - \left(\frac{1}{RC}\right)^2 = 0$

$$R = R_c = \frac{1}{2}\sqrt{\frac{L}{C}} \quad = \text{Critical Resistance.}$$

High Resistance switching :

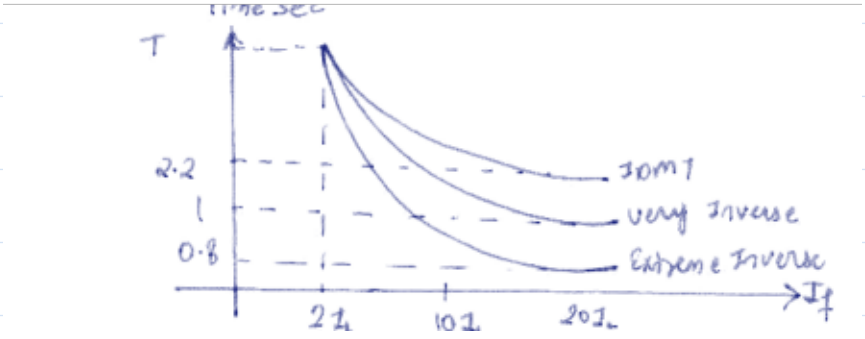
By increasing the distance between fixed and moving contact, length of the arc is increased so that resistance of arc is increased. This principle is used in high resistance switching where natural current zero is not available. It is practically preferred in DC current interruption of low magnitude.



Fixed contact (FC) and moving contact (MC) separating: arc length grows, R = rho l / A increases

Types of Relays :

* Principle : Electro magnetic, Thermal, Static
* Equipment protected : Gen, T/F, T/L, Bus bar, Motor.
* Sensing - O/C, OV, OF (overflux), E/F (Earth fault), Differential, Distance, Directional, Buchholz.
* Speed : IDMT, VI, EI, Instantaneous



Time (sec) against If: IDMT, Very Inverse and Extreme Inverse characteristics with times 2.2, 1 and 0.8 sec at 10 IL